

AI Attention ? Complete Technical Analysis

1. Project Overview

The AI Attention project is a real-time computer vision system designed to monitor and quantify user attention using webcam or video input. It integrates facial landmark detection, gaze tracking, head pose estimation, and blink analysis to compute a unified Attention Score (0-100). The system is built for applications such as online learning, remote proctoring, driver alertness, and productivity monitoring.

Key features include:

- Real-time feedback via annotated video and a web dashboard (Flask and React frontends)
- Session persistence using SQLite, with full score timelines
- REST API for programmatic access to attention data
- Support for both live webcam and uploaded video files

2. Dataset Information

[!IMPORTANT]

This is an inference-only, rule-based system. No pre-compiled training dataset is used.

| Aspect | Details |

|---|---|

| Input Source | Live webcam or uploaded video files (.mp4, .avi, etc.) |

| Data Type | Raw video frames (BGR images via OpenCV) |

| Preprocessing | BGR → RGB conversion, optional mirroring for webcam |

| Face Detection | MediaPipe Face Mesh (478 landmarks) |

| Normalization | Landmarks normalized and scaled to pixel coordinates |

3. Model Architecture

[!NOTE]

No deep learning model is trained; MediaPipe Face Mesh is used for landmark detection, with geometric and algorithmic methods for feature extraction.

Face Detection: MediaPipe Face Mesh (478 landmarks)

Gaze Tracking: Iris position ratios within eye bounding box, smoothed over 5 frames

Head Pose: OpenCV solvePnP with 6 key facial landmarks, smoothed over 7 frames

Blink Detection: Eye Aspect Ratio (EAR) algorithm, 5-frame smoothing

Attention Score: Weighted sum (Gaze 40%, Head Pose 30%, Blink 30%), with EMA smoothing

4. Training Configuration

[!IMPORTANT]

No model training is performed. All thresholds and weights are set empirically.

Parameter	Value
EMA Smoothing Alpha	0.3
Gaze Weight	0.40
Head Pose Weight	0.30
Blink Weight	0.30
EAR Threshold	0.21
Drowsy Duration	2.0 seconds

5. Feature Extraction Details

- Gaze: Horizontal/vertical iris ratios, direction classification, 5-frame smoothing
- Head Pose: Yaw, pitch, roll, direction classification, 7-frame smoothing
- Blink: EAR, blink rate, drowsiness detection, 5-frame smoothing

6. Evaluation Metrics

[!NOTE]

As a rule-based system, traditional ML metrics are not used.

Metric	Description
Attention Score	0-100, EMA-smoothed
Component Scores	Gaze, Head Pose, Blink
Status	Focused / Distracted / Drowsy
Blink Rate	Blinks per minute
Session Average	Average score per session

7. Results and Performance

Real-time processing: ~30fps on webcam input

Latency: 15-30ms per frame (MediaPipe + computation)

Score Behavior: Responsive to attention changes, robust drowsiness detection, graceful handling of no-face scenarios

8. Code Structure

AI Attention/

??? app.py # Flask server, video streaming, API

??? requirements.txt # Python dependencies

??? src/ # Core modules: face_detection, gaze_tracking, head_pose, blink_detection,
attention_score, utils

??? templates/ # Flask UI

??? static/ # CSS/JS assets

??? frontend/ # React dashboard (Vite + Tailwind)

??? uploads/ # Uploaded videos

9. Libraries and Dependencies

Backend: Python 3.x, Flask, OpenCV, MediaPipe, NumPy, SQLite3, threading

Frontend: HTML/CSS/JS (Flask), React, Vite, Tailwind CSS

10. Instructions for Execution

Clone the repo and navigate to the directory

Create and activate a Python virtual environment

Install dependencies: `pip install -r requirements.txt`

Run the backend: `python app.py`

Access via browser (Flask UI or React dashboard)

Use API endpoints for programmatic access

11. Challenges Faced and Solutions

Real-time performance: Limited face detection to 1 face, optimized array handling

Score jitter: Applied EMA and moving averages

Drowsiness detection: Required 2 seconds of continuous eye closure

Thread safety: Used locks and per-request DB connections

No-face handling: Graceful score decay and initialization logic

12. Potential Improvements

Train a supervised model for attention scoring

Add emotion recognition and audio analysis

Support multi-face and multi-modal input

Use WebSockets for real-time dashboard updates

Enhance robustness to lighting, occlusion, and adversarial scenarios

Summary

AI Attention is a robust, real-time system for attention monitoring, combining classical computer vision, signal processing, and modern web technologies. It is production-ready, extensible, and suitable for a wide range of attention-sensitive applications.